Problem Statement: Santander CIO Directive

Regulatory context & scope: Santander is mandated to comply with the European **BCBS 239** standard across data uses with the highest regulatory criticality—including FINREP, COREP, Pillar III, risk reports, and Board reporting. This requires integrating strict controls across Data Governance, Data Lineage, Data Quality, Metadata Management, and Data Cataloging.

Operational bottleneck: Compliance requires tracing information from regulatory reports all the way back to transactional core banking systems through data lakes. While end-to-end mapping and data dictionaries were created manually for Consumer, expanding field- and table-level lineage across all countries and headquarters manually is prohibitively expensive and unsustainable.

Current setup & goal: Within Openbank, Santander currently uses **DataHub** as its repository and catalog, extending it where needed. The CIO seeks to evaluate management experience and technical capabilities—specifically exploring how **AI can automatically discover field- and table-level relationships** across software assets to enhance their current DataHub setup.

1. Why Off-the-Shelf Tools (Informatica, Collibra, Standalone DataHub) Struggle Across Banking Systems

Leading catalog products like Informatica EDC, Collibra, and standard open-source DataHub work well in modern data stacks, but they hit walls when tasked with field-level lineage across multi-country banking environments:

- **Opaque Legacy Platforms:** Standard tools rely on pre-built connectors for modern cloud databases and ETL frameworks. They treat Mainframe COBOL,

DB2, PL/SQL stored procedures, and complex SAS scripts as black boxes, dropping field lineage.

- **Dynamic Runtime Transformations:** Legacy banking systems rarely use simple SQL. They rely heavily on dynamic variables, conditional business rules, and nested stored procedures that static metadata scanners fail to interpret.
- **Schema Drift Across Handshakes:** Moving data from core transactional systems through data lakes to final regulatory reports involves field renaming, aggregation, and custom script handshakes where standard scanners lose track of the line.
- **Repository vs. Automation Gap:** DataHub serves as an excellent metadata catalog and visualization layer, but it lacks out-of-the-box automated code parsers for legacy banking codebases.

2. Market Landscape: Existing Products vs. Santander's Requirements

There is **no single out-of-the-box product on the market** that packages this exact end-to-end combination—legacy banking code parsing + DataHub integration + BCBS 239 executive governance/remediation—into a plug-and-play solution. Existing vendors only solve partial pieces of the puzzle:

A. Specialized Code Lineage Engines (MANTA, Solidatus, Octopai) Partial Match

- **What they do:** Focus heavily on automated code parsing (SQL, stored procedures, COBOL, SAS) and export their graphs via OpenLineage or custom APIs.
- **Why they fall short for Santander:** They are sold as standalone, expensive SaaS platforms. Pitching them forces Santander to buy a second catalog tool on top of DataHub, causing platform redundancy. They also lack built-in executive task remediation and BCBS 239 CDE workflow controls out of the box.

B. DataHub Cloud / Acryl Data Partial Match

- **What they do:** Offers automated SQL parsing and lineage for modern cloud platforms (Snowflake, BigQuery, Databricks, dbt).

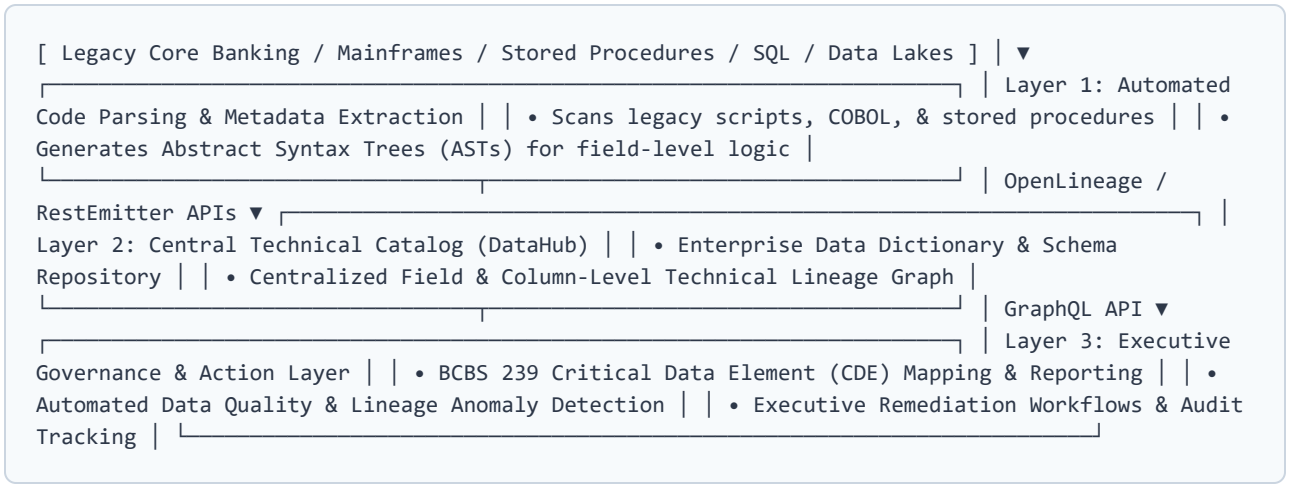
- **Why they fall short for Santander:** Native ingestion framework targets modern cloud stacks and does not natively parse legacy mainframe COBOL, complex PL/SQL packages, or custom financial report macros out of the box—which is the exact gap Openbank is facing.

C. Legacy System Modernization Vendors (Metaware, Heirloom Computing) Niche Focus

- **What they do:** Tools designed to parse legacy COBOL/Mainframe code into Abstract Syntax Trees (ASTs) for cloud migration.
- **Why they fall short for Santander:** Built for code migration rather than metadata cataloging or BCBS 239 regulatory reporting. They don't output OpenLineage graph entities to DataHub without custom wrapper development.

3. Proposed Solution Architecture: DataHub-Centric Extension

Rather than replacing DataHub or asking Openbank to start over, our proposed architecture builds directly on their existing DataHub investment. We insert an automated AI parsing layer upstream to extract field-level lineage, and link an executive governance layer downstream to manage compliance and remediation.



4. Breakdown of the Architecture Layers

Layer 1: Automated Code Parsing & Extraction

- **Automated Legacy Discovery:** Uses automated code parsers to perform static code analysis on legacy banking assets (COBOL, stored procedures, SAS, ETL jobs).
- **Field-to-Field Traceability:** Converts dynamic code and formulas into structured Abstract Syntax Trees (ASTs), building precise column-level dependencies without manual spreadsheet mapping.

Layer 2: Enterprise Catalog Repository (DataHub)

- **Single Source of Technical Truth:** Preserves Openbank's DataHub setup as the core catalog, data dictionary, and metadata graph repository.
- **Standard API Ingestion:** Ingests extracted lineage directly into DataHub using native OpenLineage and REST APIs, requiring zero changes to DataHub's core design.

Layer 3: Business Governance & Regulatory Control

- **Regulatory Mapping:** Links DataHub's technical field lineage to BCBS 239 Critical Data Elements (CDEs) across FINREP, COREP, and Board reporting.
- **Data Quality & Anomaly Monitoring:** Continuously monitors lineage paths for schema shifts, data degradation, or calculation breaks.
- **Executive Task Management & Remediation:** Automatically assigns corrective tasks to designated data stewards when a lineage or quality issue is detected, keeping an immutable audit trail for regulators.

5. Executive Positioning for Santander

> ***Proposed Position:*** *"Off-the-shelf lineage tools like MANTA require buying another expensive platform, while DataHub's default connectors struggle with legacy banking code. We bridge this exact gap. We deploy automated code parsers to scan your legacy core banking code and push field-level dependencies straight into DataHub without new software lock-in. Then, we layer executive governance and automated monitoring on top of DataHub—turning raw technical metadata into BCBS 239 audit readiness, proactive quality alerts, and clear accountability."*

6. Key Business Outcomes

1. **Drastic Effort Reduction:** Cuts manual field-mapping effort by 70–80% across country units and headquarters aggregation.
2. **Protects DataHub Investment:** Maximizes Openbank's existing platform work while closing its technical legacy gaps.
3. **Audit-Ready Compliance:** Delivers complete, field-to-field traceability from core banking databases to board reports, meeting European Banking Authority expectations.